GGG-S-121E
June 10, 1974
SUPERSEDING
Fed. Spec. GGG-S-121d
March 23, 1959 and
Amendment-5
July 14, 1965
(See section 6)

FEDERAL SPECIFICATION

SCREWDRIVER AND SCREW STARTER, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers screwdrivers intended for manual operation in driving or removing screws with single or cross recesses. Screw starters are used to start or remove screws primarily in space-restricted areas. The screwdrivers and screw starters shall be of the types normally used by cabinet makers, laborers, sheet metal workers, production workers, mechanics, and electrical repairmen.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, designs, and styles. Screwdrivers and screw starters shall be of the following types, classes, designs, and styles as specified (see 6.1). (Note: See 6.2 for deleted items.)

Type I - Flat tip.
  Class 1 - Cabinet, straight sides.
    Design A - Plain.
  Class 3 - Close quarter.
  Class 4 - Heavy duty, flared sides.
    Style 2 - Steel handle with wood inserts.
  Class 5 - General purpose, flared sides.
    Design A - Plain, medium-heavy duty.
    Design B - Bolster, forged, with hexagon-wrench grip.
  Class 6 - Flared sides with screw starter, external gripping.
  Class 7 - Pocket, removable blade.
  Class 8 - Pocket, clip.
    Style 1 - Magnetized tip.
    Style 2 - Nonmagnetized tip.

Type III - Screw starters, hand.

Type IV - Offset.
  Class 1 - Double offset, single flat tip each end.
  Class 2 - Four-way offset, double flat tip each end.
  Class 3 - Phillips cross tip each end.
    Style 1 - Double offset.
    Style 2 - Single offset.

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Type V - Ratchet.
  Class 1 - Common, straight.
  Class 2 - Double spiral, automatic return.
    Style 1 - Light duty.
    Style 2 - Medium duty.
    Style 3 - Heavy duty.

Type VI - Cross tip, straight.
  Class 1 - Phillips.
  Class 2 - Reed and Prince.
  Class 3 - Phillips, with screw starter, external gripping, with cushion grip.

Type VII - Screwdriver, spark testing.

Type VIII - Bit set and handle.
  Class 1 - Plastic handle with clutch drive.
  Class 2 - Bit, Phillips cross tip.
  Class 3 - Bit, cabinet-type tip.
  Class 4 - Extension, bit.

Type IX - Flat tip - cross tip, close quarter.

Type X - Clutch tip.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

VV-B-680 - Brake Fluid, Automotive.
PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI), Inc., Standards:

B18.6.3 - Slotted and Recessed Head Machine Screws and Machine Screw Nuts (ISO R272).
B46.1 - Surface Texture.
B107.4 - Driving and Spindle Ends for Portable Hand, Air, and Electric Tools.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
D2240-68 - Standard Method of Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of screwdrivers which are otherwise in accordance with the requirements of this specification.

3.2 Identification marking. Each item shall be marked in a plain and permanent manner with the manufacturer's name or with a U. S. registered trademark so that the source of supply may be readily identified. Screwdrivers shall not be marked with wording such as "shockproof" which may imply to the user that the handle provides complete electrical protection.

3.3 Test loads. When tested as specified in 4.4.2 through 4.4.4, the tools shall withstand the test loads of the applicable tables without breakage or permanent visible distortion.

3.4 Blades. Blades shall be of steel, properly heat treated to withstand the tests specified herein. All four sides, and the end of flat tip blades, shall be symmetrical. The tip thickness shall be uniform across its entire width within 0.003 inch. The tang end of the blade shall be held securely in the handle. The blades shall be free from scale, seams, laps, and cracks.

3.4.1 Finish. The blades shall have a smooth, bright finish that will produce a 125 microinch roughness height rating or better (as determined in accordance with ANSI B46.1). The blades shall be treated in a manner to prevent rust or corrosion. Plating, if any, shall be adherent. There shall be no evidence of plating peeling, or chipping during testing procedures specified in 4.4.2 through 4.4.10, with exception of 4.4.5. Cadmium plating will not be acceptable.

3.4.2 Hardness. Unless otherwise specified herein, the tip portion of the screwdrivers, or the entire blade if it is the manufacturer's standard practice, shall be hardened to not less than 48 nor more than 58 on the Rockwell C scale when tested as specified in 4.4.1.

3.5 Plastic handles. The handles shall be of plastic capable of withstanding the applicable test requirements of 4.4.6. They shall be transparent amber, or yellow in color, and chemically polished or suitably finished to provide a comfortable grip. They shall be free from rough edges, sharp corners, or tool marks that affect the appearance of the tool. Cross sections shall be essentially square, hexagonal, triangular, round, or octagonal. Square, hexagonal, and triangular handles shall be rounded on the edges to a minimum radius equal to the distance across the flats divided by four. Round and octagonal handles shall be fluted with four to six major flutes for a length at least 65 percent of the handle length. The size of the flute and the major periphery shall make an angle of not less than 90°, and the juncture shall be broken to 1/64 radius minimum. The handle butt shall be well rounded to a minimum radius to 1/8 of the major cross-sectional dimension and the run out of the radius with the major periphery shall not exceed an angle of 15°. The handle volume shall be at least 53 percent of the major cross-sectional dimension times the length for standard handles, and 50 percent for stubby handles.

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3.6 Handle with 3/8-inch female drive insert. When specified (see 6.1), handles shall have a metal 3/8-inch square female drive insert of a design suitable for accepting a corresponding 3/8-inch square male drive tang in accordance with table 9, ANSI B107.4. The insert shall be flush with the butt end of the screwdriver handle. If the insert is made of ferrous material, it shall be treated to prevent corrosion. The insert shall meet the test requirements in 4.4.9. When handles with inserts are furnished, the diameter of the handle shall not be less than 1-1/32 inch. One or more faces of the female drive opening shall be drilled or recessed so that it will engage the spring-loaded steel ball, or steel plunger on the corresponding male drive.

3.7 Cushion grip. When specified (see 6.1), handles shall be furnished with cushion grips. The grips shall be of rubber or vinyl capable of meeting the tests of 4.4.6. The length shall be at least 60 percent of the handle length, and there shall be no detectable slippage between the handle and the cushion grip under a normal usage test. The original hardness shall be Shore A 60 to 75 when tested in accordance with ASTM D2240-68, and the subsequent change after the brake fluid and oil test (see 4.4.6), shall not be greater than 80 Shore A. The cushioned handles shall meet the dimensional requirements for corresponding conventional type handles for each type, class, and style (see applicable tables) except that an additional 7/8 inch plus tolerance will be permitted on the handle length, and 3/16 inch plus tolerance will be permitted on the diameter of the handle. The fluting requirements for handles do not apply when cushion grips are supplied.

3.8 Type I, flat tip. The blades of flat tip screwdrivers shall be in accordance with 3.4, 3.4.1, and 3.4.2. The included angle between bearing surfaces shall be 10° maximum. The extreme tip width shall be perpendicular to the blade axis to within 2°, and the tip thickness to within 6°. The tip thickness shall be uniform to within 0.003 inch.

3.8.1 Type I, class 1, design A. The flat tip, plain cabinet type screwdrivers shall consist essentially of a plastic handle and a blade. The entire length of the blade shall be capable of entering a round hole no larger in diameter than the maximum allowable tip width dimension, plus 1/32 inch. The screwdriver shall be similar to figure 1 and conform to table I.

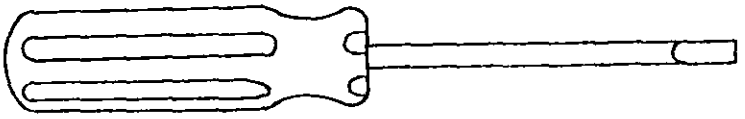


FIGURE 1. Type I, class 1, design A screwdriver.

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TABLE I. Type I, class 1, design A screwdrivers

Blade			Handle				
Exposed length ±1/4	Tip width at end ±1/64	Thickness of tip at end ±0.004	Diameter or distance across corners (Min.)	Length (Min.)	Torsional moment (Min.)	Bending moment (Min.)	Test block (see table XXVIII)
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	Inch-Pounds	
1-1/2 2-1/2 3	9/64	0.020	3/4	1-3/4	8	60	B
4 5 8 10 12	9/64	.020	3/4	2-5/8	8	60	B
1-1/2	3/16	.030	15/16	2-5/8	25	100	D
3 5 6 8	3/16	.030	15/16	3-1/4	25	100	D
10 12 15	7/32	.032	1	3-1/2	30	130	E
6 8 10 18	1/4	.037	1	3-1/2	40	180	F

3.8.2 Type I, class 3, close quarter. Type I, class 3, close quarter screwdrivers shall be suitable for working in close quarters, and shall consist essentially of a blade and a plastic handle. The handle shall be in accordance with 3.5. The screwdrivers shall be similar to figure 2 and conform to table II.

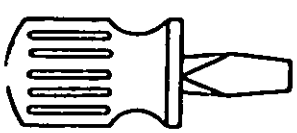


FIGURE 2. Type I, class 3 screwdriver, flat tip, close quarter.

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TABLE II. Type I, class 3 screwdriver, flat tip, close quarter

Blade			Handle		Torsional moment (Min.)	Test block (see table XXVIII)
Exposed length ±1/4	Tip width at end ±1/64	Thickness of tip at end ±0.004	Diameter or distance across corners (Min.)	Length (Min.)		
Inches	Inch	Inch	Inches	Inches	Inch-pounds	
1	7/32	0.032	1-1/8	1-1/2	35	E
1-1/2	1/4	.037	1-1/8	1-1/2	60	F
1-3/4	5/16	.042	1-1/8	1-1/2	60	G

3.8.3 Type I, class 4, style 2, heavy duty, flared sides. The blade and handle frame shall be forged in one piece from high-grade steel. A hardwood insert shall be secured to each side of the handle frame with not less than two rivets. The rivet heads and outer surface of the inserts shall be finished flush with the outer surface of the handle frame. The cross-sectional shape of the handle shall be approximately elliptical, smoothly and uniformly finished with the greater diameter of the ellipse being in alignment with the narrow edges of the flat tip of the blade. The inserts shall be protected at both ends by the metal portion of the handle. The wood shall be clear, straight-grained, white ash, beech, birch, hickory, hard maple, or pecan. Moisture content shall not exceed 12 percent. Handles shall be free of defects except that mineral streaks and sound, tight knots of not over 1/8 inch diameter will be allowed. The screwdriver shall be similar to figure 3 and conform to table III.

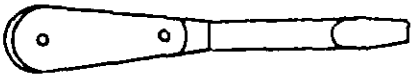


Figure 3. Type I, class 4, style 2, heavy duty, flared sides.

TABLE III. Type I, class 4, style 2, heavy duty, flared sides

Blade			Handle		Torsional moment (Min.)	Bending moment (Min.)	Test block (see table XXVIII)
Exposed length ±1/4	Tip width at end +1/16 -1/32	Thickness of tip ±0.004	Width at largest section +1/4 -1/16	Length (Min.)			
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	Inch-Pounds	
4 1/	5/16	0.042	1-1/8	4	80	700	G
6	3/8	.050	1-3/8	4	140	1,000	H
8	7/16	.058	1-1/2	5	200	1,200	I
10	7/16	.058	1-1/2	5	200	1,200	I
5 2/	1/2	.070	1-3/8	4	250	1,400	J

1/ Furnished with shot-blast finish.

2/ Extra heavy duty. Furnished with square blade, and shot-blast finish.

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3.8.4 Type I, class 5, designs A and B, general purpose, flared sides. Type I, class 5 screwdrivers shall consist essentially of a blade and a handle. The handle shall be in accordance with 3.5. Design A blades shall be of the plain, round blade type similar to figure 1. Design B blades shall be similar to figure 1 except it shall have a plain-type bolster on sizes 6 inches and smaller, or hexagon-wrench type bolster on larger sizes forged integral with the blade. The outside diameter of the bolster shall be at least 50 percent larger than the diameter of the blade. When the blade and handle are assembled, the bolster shall fit tightly against the handle. The screwdrivers shall conform to the requirements shown in table IV for the size specified (see 6.1).

TABLE IV. Type I, class 5, designs A and B, general purpose, flared sides

Blade			Handle		Torsional moment (Min.)	Bending moment (Min.)	Test block (see table XVIII)
Exposed length ±1/4	Tip width at end ±1/32 -1/64	Thickness of tip at end ±0.004	Diameter or distance across corners (Min.)	Length (Min.)			
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	Inch-Pounds	
3	7/32	0.032	15/16	3-1/4	30	175	E
4	1/4	.037	1	3-1/2	40	350	F
6	3/16	.042	1-1/8	3-1/2	60	700	G
6	5/16	.042	1-1/8	3-1/2	60	700	G
8	5/16	.042	1-1/8	3-1/2	60	700	G
10	3/8	.050	1-1/8	4-5/8	140	1,000	H
12							
18	1/2	.070	1-1/2	4-5/8	200	1,000	J

3.8.5 Type I, class 6, flared sides with screw starter, external gripping. Type I, class 6 screwdrivers shall be suitable for holding screws when starting or removing screws from restricted places, and shall be suitable for normal screwdriver work. They shall consist essentially of a blade, screw-gripping mechanism, and a handle. The screw gripper shall be of the spring-action type capable of holding any size screw for which the point thickness of the blade is suited, and passing the tests of 4.4.8. The gripper shall slide the full length of the blade, and shall have sufficient friction to remain in any position on the shank. The handle shall conform to 3.5. The screwdriver shall be similar to figure 4, and conform to table V.

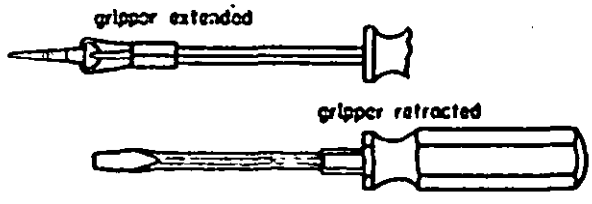


FIGURE 4. Type I, class 6, flared sides with screw starter, external gripping.

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TABLE V. Type I, class 6, flared sides with screw starter, external gripping

Blade			Handle		Torsional moment (Min.)	Bending moment (Min.)	Test block (see table XXVIII)
Exposed length ±1/4	Tip width at end +1/32 -1/64	Thickness of tip at end ±0.004	Diameter or distance across corners (Min.)	Length (Min.)			
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	Inch-Pounds	
2	5/32	0.025	5/8	2	6	80	C
4 6 10	5/32	.025	5/8	2-5/8	12	80	C
4	7/32	.032	15/16	3-1/4	30	175	E
8	5/16	.042	1-1/8	3-1/2	80	500	C

3.8.6 Type I, class 7, pocket, removable blade. Type I, class 7 screwdrivers shall be of steel and shall consist essentially of a handle, blade, and a chucking device. The body of the handle shall be of a round cross-sectional shape, and shall be neatly knurled. External surfaces shall be protected by nickel plating or other suitable and equally effective means. The upper portion of the handle shall be provided with a knob at least 1/8 inch larger than the diameter of the handle. The handle shall be provided with a chuck so designed that the shank of the blade may be readily locked in the handle. The design shall also permit the blade to be inverted and inserted in the handle when not in use. Additional blades shall be furnished when specified (see 6.1). The screwdrivers shall be similar to figure 5 and shall conform to table VI.

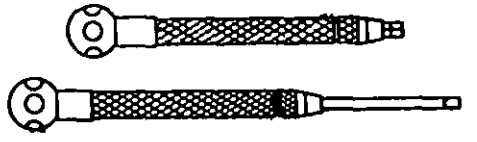


FIGURE 5. Type I, class 7, pocket, removable blade.

TABLE VI. Type I, class 7, pocket, removable blade

Size number	Blade			Body diameter ±1/32	Closed length (Min.)
	Tip width at end +0.015 -0.010	Thickness of tip at end +0.003 -0.002	Length under chuck ±1/4		
	Inch	Inch	Inches	Inch	Inches
1	0.090	0.010	1	1/4	3
2	.156	.014	2	3/8	4

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3.8.7 Type I, class 8, styles 1 and 2 pocket clip. Type 1, class 8 screwdrivers shall consist of a blade, handle, and a pocket clip. The clip shall be of corrosion-resistant metal, or treated to resist corrosion, and shall be either a slip-on type or permanently secured to the screwdriver handle. The movable end of the pocket clip shall have a rounded section which shall contact the handle within 0.006 inch. The clip shall have sufficient spring action to return to its normal position after a 3/16-inch diameter rod is passed between the handle and the clip end. Blades shall be style 1 (magnetized) or style 2 (nonmagnetized), as specified (see 6.1) and shall pass the tests of 4.4.11. Handles shall be in accordance with 3.5. The screwdrivers shall be similar to figure 6 and conform to table VII.

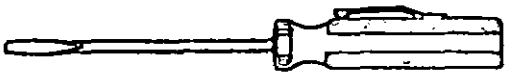


FIGURE 6. Type I, class 8, styles 1 and 2, pocket clip.

TABLE VII. Type I, class 8, styles 1 and 2, pocket clip

Blade			Handle		Torsional moment (Min.)	Test block (see table XXVIII)
Exposed length ±1/4	Tip width at end ±1/64	Thickness of tip at end ±0.004	Diameter or distance across corners (Min.)	Length (Min.)		
Inches	Inch	Inch	Inch	Inches	Inch-Pounds	
2	3/32	0.015	7/16	1-3/4	4	A
2 3 4	1/8	.020	1/2	1-3/4	6	B

3.9 Type III, screw starters, hand. Type III screw starters shall consist essentially of a blade, screw-slot gripping mechanism, and a handle. The screw starters shall be for use on straight-slotted screws. They shall grip the screw internally within the slot of the screw head by a wedge action between the blade and the screw slot. They shall be suitable for starting or removing screws from restricted places and be capable of passing the tests of 4.4.8. The tip and the part of the gripper which engages the screw slot shall show a hardness of not less than 45 nor more than 58 on the Rockwell C scale. The reading shall be taken as near the tip end as practicable. Handles shall be of plastic or metal.

3.9.1 Plastic handle. Plastic handles shall be in accordance with 3.5.

3.9.2 Metal handle. Metal handles shall be of aluminum alloy, neatly knurled, and shall be integral with or securely attached to the blade body or shank.

3.9.3 Type III screw starters shall conform to the dimensions specified in table VIII.

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TABLE VIII. Type III screw starter, hand

Blade			Handle	
Exposed length ±1/4	Tip width at end +1/16 -1/64	Thickness of tip at end ±0.004	Diameter or distance across corners (Min.)	Length (Min.)
Inches	Inch	Inch	Inch	Inches
1-1/4	3/16	0.030	15/16	2-5/8
2-3/4	9/32	.032	15/16	2-5/8
4	1/4	.037	1	3-1/2
6	1/4	.037	1	3-1/2
6	9/32	.032	1	3-1/2
8	3/16	.030	1	3-1/2

3.10 Type IV, offset.

3.10.1 Class 1, double offset, flat tip each end. Class 1 screwdrivers shall be of one-piece steel construction. The flat tips shall be at right angles to each other, and conform to 3.4. The cross-sectional shape of the body shall be octagon, hexagon, or square. When hexagon or square-shaped stock is used, the edges along the entire body length shall be well rounded to provide a comfortable grip. In lieu of octagon, hexagon, or square-shaped body, round will be acceptable, provided it is well-knurled or has a coarse shot-blast finish for firm gripping. The screwdrivers shall be similar to figure 7 and conform to table IX.

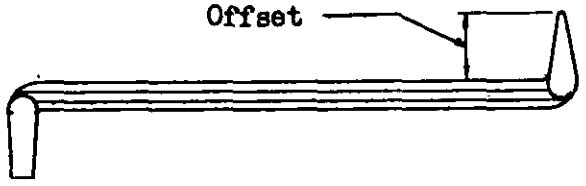


FIGURE 7. Type IV, class 1, double offset.

TABLE IX. Type IV, class 1, double offset

Tip width at end +1/32 -1/64	Offset ±1/4	Thickness of tip at end ±0.004	Length ±1/2	Diameter or distance across flats of body ±1/32	Torsional moment (Min.)
Inch	Inch	Inch	Inches	Inch	Inch-Pounds
5/32	1/2	0.025	3	5/32	12
1/4	1/2	.037	4-1/4	7/32	40
5/16	5/8	.042	5	9/32	80
3/8	3/4	.050	6	11/32	140
1/2	7/8	.070	6	13/32	250
3/4	1	.130	7-1/2	1/2	400

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3.10.2 Type IV, class 2, four-way offset, flat tips each end. Class 2 screwdrivers shall be of steel, and may be of one-piece construction, or have blades securely fastened into the ends. The tips shall be arranged in such a manner that eight positions in 360° are available to engage screw slots. There shall be four blades, two at each end, protruding in opposite directions with one blade being in line with the longitudinal axis of the body and the other blade at right angles to the axis. The two blades at the other end of the body shall be parallel to each other and set at 45° to the axis of the body. The axis of the blades at both ends shall be parallel. The tips and hardness shall be in accordance with the applicable requirements of 3.4 and 3.4.2 and the blade tips shall be flared. The cross-sectional shape of the body shall conform to 3.10.1. The screwdriver shall be similar to figure 8 and conform to table X.

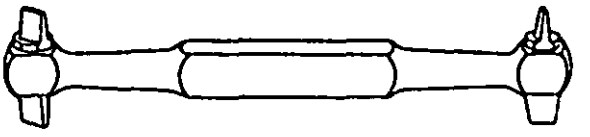


FIGURE 8. Type IV, class 2, four-way offset.

TABLE X. Type IV, class 2, four-way offset

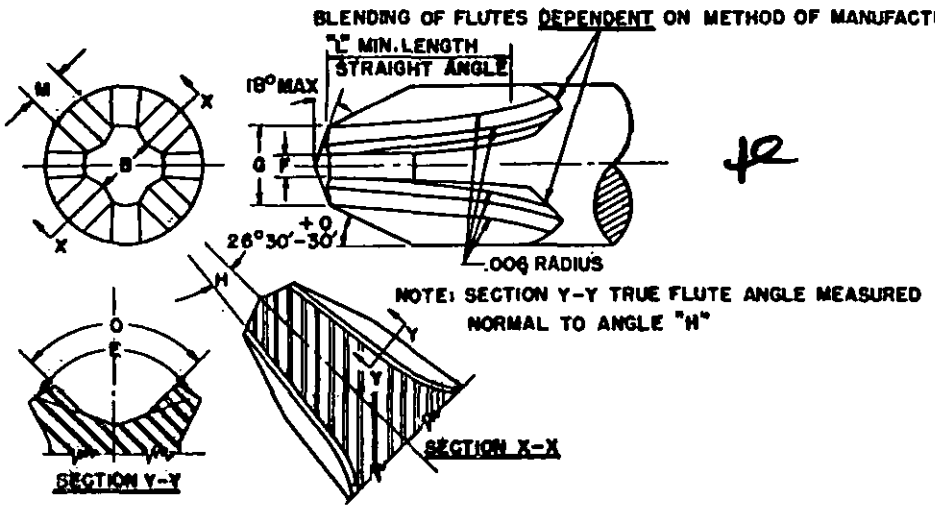
Tip width at end ±1/64	Offset +1/4 -1/16	Thickness of tip at end ±0.004	Length +3/8 -1/4	Diameter or distance across flats of body +3/16 -1/16	Torsional moment (Min.)
Inch	Inch	Inch	Inches	Inch	Inch-Pounds
1/4	5/16	0.037	5-1/4	5/16	40

3.10.3 Type IV, class 3, style 1, double offset, Phillips cross tip each end. The class 3 screwdrivers shall have Phillips-type points formed on opposite sides of the tool. They shall be of one-piece steel construction and capable of withstanding the test specified in 4.4.1 through 4.4.2.1. Each cross tip blade shall show a hardness of not less than 52 nor more than 60 on the Rockwell C scale. The reading shall be taken as near the tip end as practicable.

3.10.3.1 Phillips screwdriver points shall conform to the dimensions shown on figure 9. Conformance to the required dimensions shall be determined by the use of gages manufactured in accordance with the applicable dimensions shown on figure 10. The wing tips shown on figure 9 for dimension "G" shall fall within the step on the gage designated as the "C" dimension shown on figure 10. In addition, the driver and bit points shall fit firmly within the gage. When plating is used for Phillips screwdrivers, the ring gages shall be used prior to plating, or the plating shall be removed for ring gaging purposes. In addition to the marking requirements of 3.2, the screwdrivers shall be marked with the name "Phillips" and the size number of point(s). The screwdrivers shall be similar to figure 11 and conform to table XI.

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PHILLIPS SCREWDRIVER & BIT POINT DIMENSIONS
(NO. 0,1,2,3 & 4 POINT SIZES)



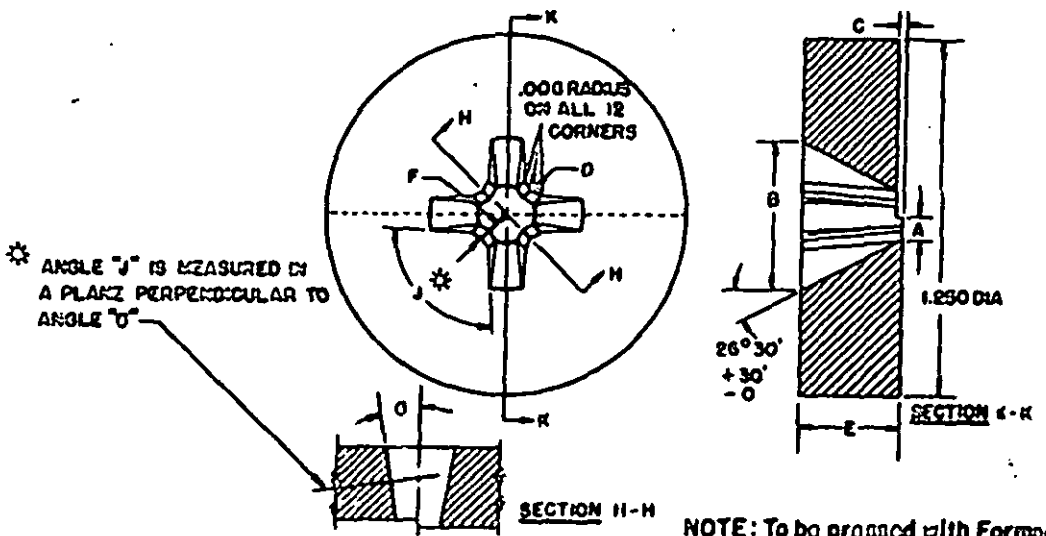
NOTE:
FOR NO. 0 SIZE
E EQUALS 0.004
RADIUS APPROX.

POINT SIZE	±0.001 B	+ 30' - 0 E	±0.001 G	+ 0 - 30' H	F	+ 30' - 0 O	±0.001 M	L
NUMBER	INCH	DEGREE	INCH	DEGREE	INCH	DEGREE	INCH	INCH
0	0.023	SEE NOTE	0.032	7°	0.012 0.014	92	0.0078	1/8
1	0.0394	138	0.050	7°	0.018 0.020	92	0.0202	1/8
2	0.0606	140	0.090	5°45'	0.023 0.028	92	0.0434	3/16
3	0.0983	146	0.150	5°45'	0.027 0.033	92	0.0826	9/32
4	0.1407	153	0.200	7°	0.040 0.045	92	0.1078	11/32

FIGURE 9. Phillips screwdriver point dimensions and requirements.

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RING GAGE FOR PHILLIPS SCREWDRIVER & BIT POINTS



NOTE: To be produced with Formed Ground Master Punch.

POINT SIZE	± 0.0002 A	(REF) B	± 0.001 C	(REF) D	$+0\ -\frac{1}{64}$ E	± 0.0002 F	+15' -0 G	J
NUMBER	INCH	INCH	INCH	INCH	INCH	INCH	DEGREE	DEGREE
0	0.014	0.123	0.010	0.0268	$\frac{8}{32}$	0.0114	7°	92
1	0.0255	0.1785	0.010	0.0634	$\frac{1}{6}$	0.0194	7°	92
2	0.0455	0.2800	0.010	0.0971	$\frac{3}{10}$	0.0303	6°45'	92
3	0.0755	0.4348	0.010	0.1538	$\frac{9}{32}$	0.0496	6°45'	92
4	0.1005	0.5471	0.010	0.2083	$\frac{11}{32}$	0.0711	7°	92

FIGURE 10. Ring gages for inspecting Phillips screwdriver and bit points (see figure 11).

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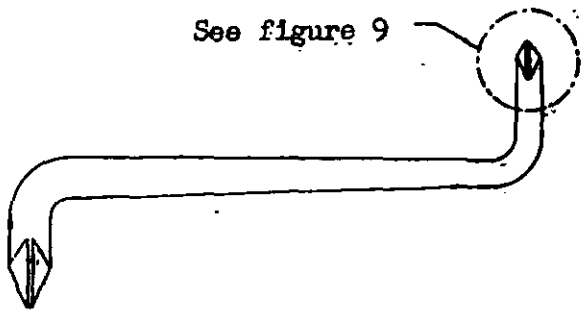


FIGURE 11. Type IV, class 3, style 1, double offset, Phillips cross tip.

TABLE XI. Type IV, class 3, style 1 screwdriver, double offset, Phillips cross tip each end

Point size		Length of each offset		Overall length ±1/4	Diameter of shank (Max.)		Torsional moment	
Small end	Large end	Small end ±3/16	Large end ±3/16		Small end	Large end	Small end (Min.)	Large end (Min.)
Number	Number	Inches	Inches	Inches	Inch	Inch	Inch-Pounds	Inch-Pounds
1	2	3/4	7/8	4-3/4	0.197	0.260	25	60
2	3	7/8	1-1/16	5-1/2	.260	.322	60	150
3	4	1-1/16	1-5/16	6	.322	.385	150	200

3.10.4 Type IV, class 3, style 2, single offset. The class 3, style 2 screwdrivers shall have only one end offset and the points on each end shall be the same size. Style 2 screwdrivers shall be similar to figure 12 and conform to the requirements in table XII.

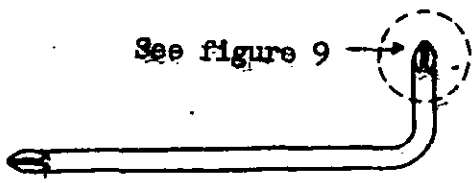


FIGURE 12. Type IV, class 3, style 2 screwdriver, single offset, Phillips cross tip each end.

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TABLE XII. Type IV, class 3, style 2 screwdriver, single offset, Phillips cross tip each end

Point size	Length of offset ±1/16	Overall length ±1/4	Diameter of shank ±0.010	Torsional moment (Min.)
Number	Inches	Inches	Inch	Inch-Pounds
2	1-1/8	4	1/4	60

3.11 Type V, ratchet.

3.11.1 Class 1, common, straight. The class 1 screwdrivers shall be those known commercially as ratchet screwdrivers. They shall consist essentially of a blade, handle, ratchet mechanism, and a shifter. The blade shall conform to 3.4, 3.4.1, and 3.4.2. The gripping portion of the handle shall be smoothly finished and suitably shaped to provide a good handgrip. The ratchet mechanism shall be of a design to permit the screwdriver to operate as a right- or left-handed ratchet screwdriver, or as a rigid tool. The screwdriver shall be similar to figure 13 and conform to table XIII.



FIGURE 13. Type V, class 1, common, straight.

TABLE XIII. Type V, class 1, common, straight

Blade			Handle		Torsional moment (Min.)	Test block (see table XIVIII)
Exposed length ±1/4	Tip width at end +1/32 -1/64	Thickness of tip at end ±0.004	Diameter or distance across corners (Min.)	Length (Min.)		
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	
3	5/16	0.042	1-1/4	3-1/2	60	C
6	5/16	.042	1-1/4	4	60	C

3.11.2 Type V, class 2, ratchet, double spiral, automatic return. Class 2 screwdrivers shall consist essentially of a handle, double spiral spindle, a chuck suitable for holding screwdriver bits specified herein, spiral nuts, a shifter, three flat tip bits (see 3.11.2.1), and such other parts as may be necessary for proper functioning. The screwdriver shall operate by a reciprocating movement of the handle, a spring being utilized to automatically return the handle to the extended position. It shall be suitable as a right- or left-handed ratchet screwdriver, or as a rigid screwdriver. The gripping portion of the handle shall be of plastic in accordance with 3.11.1. The spindle shall have a right-hand and left-hand spiral, each of which will produce one revolution when the handle travels the following distances: ±1/4 inch; 1-1/2 inches for the light duty; 2 inches for the medium duty; and 3 inches for the heavy duty. The entire surfaces of the spindle and the spiral grooves shall be uniform and smoothly finished. The screwdrivers shall retain the bits described herein, be similar to figure 14 and conform to table XIV.

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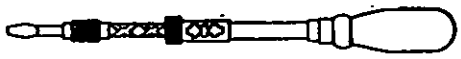


FIGURE 14. Type V, class 2, ratchet, double spiral.

TABLE XIV. Type V, class 2, ratchet, double spiral

Style	Length closed (Max.)	Length of stroke (Min.)	Handle diameter ±1/32	Compression test load (with bit inserted) (Min.)
	Inches	Inches	Inches	Pounds
Style 1, light duty	10-3/8	3-1/8	1-1/4	65
Style 2, medium duty	15-1/8	5	1-3/8	100
Style 3, heavy duty	19-3/4	8	1-1/2	150

3.11.2.1 Flat tip bits. The bits shall conform to 3.4, 3.4.1, and 3.4.2. Three bits (one of each bit number specified in table XVII for each style 1, 2, and 3 screwdriver) shall be supplied with type V, class 2 screwdrivers. When specified (see 6.1), bits shall be supplied separately. The bits shall conform to figure 15 and tables XV and XVI.

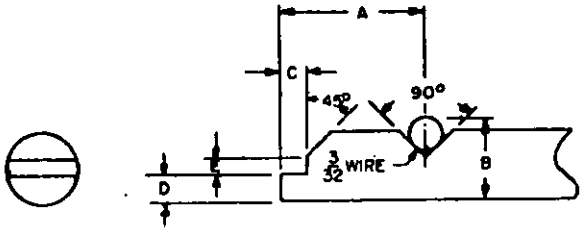


FIGURE 15. Chuck end of bits and adapter for type V, class 2, ratchet screwdriver.

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TABLE XV. Chuck end dimensions

Nominal diameter for chuck end of bits	A ±0.010	B ±0.005	C ±0.010	D ±0.004	E ±0.015
Inch	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
1/4	0.555	0.270	0.120	0.094	0.075
5/16	.650	.330	.120	.124	.085
3/8	.650	.355	.120	.138	.085

TABLE XVI. Flat tip bits for type V, class 2, ratchet screwdrivers

Style	Bit number	Bit dimensions				Torsional moment (minimum)
		Over-all length +1/16 −1/4	Width of tip at end ±0.010	Thickness of tip at end ±0.004	Diameter at chuck end +0.001 −.004	
		Inches	*Inch*	*Inch*	*Inch*	*Inch-Pounds*
Style 1, light duty	1	3	5/32	0.020	7/32	10
	2	3	3/16	.025	7/32	20
	3	3	7/32	.030	7/32	30
Style 2, medium duty	4	3-7/8	7/32	.030	9/32	30
	5	3-7/8	1/4	.035	9/32	38
	6	3-7/8	9/32	.040	9/32	50
Style 3, heavy duty	7	3-7/8	1/4	.035	5/16	38
	8	3-7/8	9/32	.040	5/16	50
	9	3-7/8	5/16	.042	5/16	70

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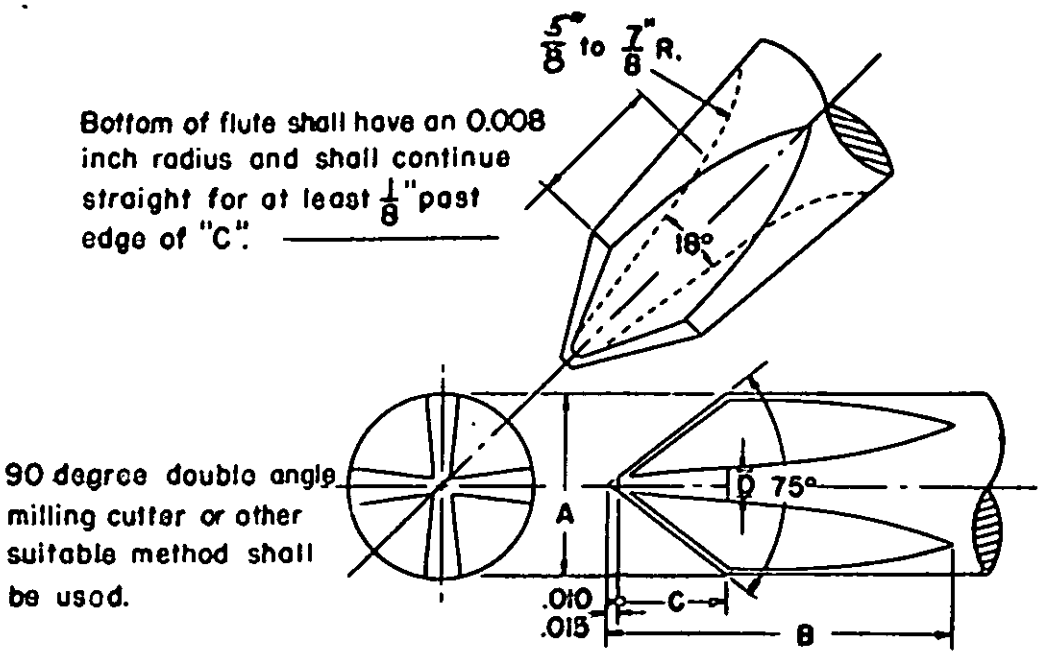
3.11.2.2 Phillips cross-point bits. When specified (see 6.1), bits with Phillips-type points shall be furnished. They shall be in accordance with the applicable requirements in 3.10.3 and 3.10.3.1. The tips shall be similar to figure 9 and conform to table XVII.

TABLE XVII. Phillips cross-point bits for type V, class 2, styles 1, 2, and 3 Screwdrivers, ratchet double spiral, automatic return, light duty, medium duty, heavy duty

Style	Bit number	Bit dimensions				Torsional moment (minimum)
		Phillips point size	Diameter at point end ±0.010	Diameter at chuck end +0.001 − .004	Over-all length ±1/4	
		Number	*Inch*	*Inch*	*Inches*	*Inch-Pounds*
Style 1, light duty	1	1	3/16	7/32	3-1/4	25
	2	2	7/32	7/32	3-1/4	60
Style 2, medium duty	3	1	3/16	9/32	3-1/4	25
	4	2	9/32	9/32	3-1/4	60
	5	3	5/16	9/32	3-1/4	150
Style 3, heavy duty	6	2	1/4	5/16	3-1/4	60
	7	3	5/16	5/16	3-1/4	150

3.11.2.3 Reed and Prince cross-point bits. When specified (see 6.1), bits with Reed and Prince type points shall be furnished. They shall be of one-piece steel construction, and capable of withstanding the test specified in 4.4.1, 4.4.2.1, and 4.4.2.2. Each cross tip blade shall show a hardness as specified in 3.10.3. The bits shall be capable of tightening and loosening screws having a Reed and Prince recess in accordance with ANSI B18.6.3. The chuck end of the bit shall conform to figure 15 and conform to table XV. When specified (see 6.1), bits shall be supplied separately. In addition to the marking requirements of 3.2, the screwdriver bits shall be permanently marked with the name "Reed and Prince". The screwdriver tip shall conform to the dimensions shown on figure 16 and be of the sizes specified in table XVIII.

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BLADE SIZE A	B	C	D
INCH	INCH	INCH	INCH
0.187	1/2	0.122	0.041
0.250	9/16	0.163	0.050
0.312	5/8	0.204	0.060
0.375	11/16	0.244	0.069

FIGURE 16. Reed and Prince screwdriver-point dimensions and requirements.

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TABLE XVIII. *Reed and Prince cross-point bits for type V, class 2, styles 1, 2, and 3 screwdrivers, ratchet, double spiral, automatic return, light duty, medium duty, heavy duty*

Style	Bit number	Bit dimensions			Torsional moment (minimum)
		Diameter		Over-all length ±¼	
		Point end ±0.010	Chuck end +0.001 − .004		
		Inch	*Inch*	*Inches*	*Inch-Pounds*
Style 1, light duty	1	7/32	7/32	3-1/4	40
Style 2, medium duty	2	9/32	9/32	3-1/4	90
Style 3, heavy duty	3	5/16	5/16	3-1/4	150

3.12 Type VI, cross tip, straight.

3.12.1 Class 1, Phillips. Class 1 screwdrivers shall have Phillips-type points as specified in 3.10.3. The blade shall be round in cross section, formed from high grade steel, hardened, and marked to conform to the requirements of 3.10.3 and 3.10.3.1, and be capable of withstanding the test specified in 4.4.2.1. The screwdrivers shall have plastic handles in accordance with 3.5, be similar to figure 17 and conform to table XIX.

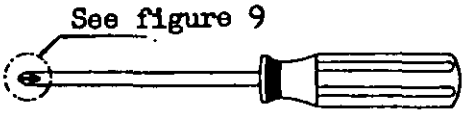


FIGURE 17. Type VI, class 1 screwdriver, cross tip, straight, Phillips.

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TABLE XIX. Type VI, class 1 screwdriver, cross tip, straight, Phillips

Blade			Handle		Torsional moment (Min.)
Exposed length ±1/4	Point size	Diameter ±0.010	Diameter (Min.)	Length (Min.)	
Inch	Number	Inch	Inches	Inches	Inch-Pounds
1-1/4	0	1/8	1/2	1-1/2	6
4	0	1/8	1/2	1-1/2	6
1 1/	1	3/16	13/16	1-1/2	25
8 2/	1	3/16	13/16	3-1/4	25
3	1	3/16	13/16	3-1/4	25
1-1/2 1/	2	1/4	7/8	1-1/2	40
4	2	1/4	7/8	3-1/2	40
8 2/	2	1/4	7/8	3-1/2	40
10	2	1/4	7/8	3-1/2	40
6	3	5/16	1-1/16	3-1/2	90
8	4	3/8	1-1/4	4-5/8	150

1/Close quarter screwdrivers.

2/Extra long blade.

3.12.2 Class 2, Reed and Prince. Type VI, class 2 screwdrivers shall have Reed and Prince type points as specified in 3.11.2.3, and conform to the dimensions on figure 16. The screwdrivers shall have plastic handles in accordance with 3.5, be similar to figure 18 and conform to the dimensions specified in table XX.

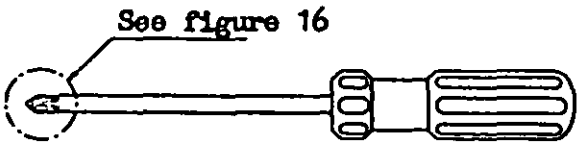


FIGURE 18. Type VI, class 2 screwdriver, cross tip, straight, Reed and Prince.

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TABLE XX. Type VI, class 2 screwdriver, cross tip, straight, Reed and Prince.

Blade		Handle		Torsional moment (Min.)
Exposed length ±1/4	Diameter ±0.010	Diameter (Min.)	Length (Min.)	
Inches	Inch	Inches	Inches	Inch-Pounds
3	3/16	15/16	3-1/4	25
10	3/16	1	3-1/2	25
4	1/4	1	3-1/2	40
6	5/16	1-1/8	3-1/2	90
15	5/16	1-1/8	4-5/8	90
8	3/8	1-1/8	4-5/8	150
1 1/	3/16	13/16	1-7/8	15
1-1/2 1/	1/4	13/16	1-7/8	20

1/ Close quarter screwdrivers.

3.12.3 Class 3, Phillips, with screw starter, external gripping, plastic handle, with cushion grip. The type VI, class 3 screwdrivers shall be in accordance with type VI, class 1, except for the following: The screwdriver handle shall be provided with a cushion grip in accordance with 3.7, and the blade and shank shall be provided with a gripping mechanism in accordance with 3.8.5. The screwdrivers shall be in accordance with the 3 inch size, Point No. 1; or 4 inch size, Point No. 2, listed in table XIX, as specified (see 6.1).

3.13 Type VII, spark testing. Type VII screwdrivers shall be suitable for testing spark plugs in automobile engines and other gasoline engines. They shall consist essentially of a blade and plastic handle with a neon tube. The blade shall withstand the hardness and torsional moment tests of 4.4.1 and 4.4.2. The handle shall not be subjected to an impact test. A pocket clip shall be furnished on the 2-1/2 inch blade length screwdriver. The clip shall be in accordance with 3.8.7. The screwdriver shall be similar to figure 19 and conform to table XXI.

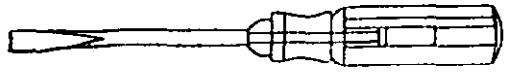


FIGURE 19. Type VII screwdriver, spark testing.

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TABLE XII. Type VII screwdriver, spark testing

Blade dimensions			Handle dimensions		Torsional moment (Min.)
Exposed length ±1/2	Diameter and width of tip at end +0.065 -0.005	Thickness of tip at end ±0.004	Diameter (Min.)	Length (Min.)	
Inches	Inch	Inch	Inch	Inches	Inch-Pounds
2-1/2	1/8	0.020	7/16	2	6
6	3/16	.030	7/8	3-1/2	20

3.13.1 Blade. The blade for type VII screwdrivers shall conform to 3.4, 3.4.1, and 3.4.2. The cross-sectional shape shall be round.

3.13.2 Handle. The handle shall conform to the applicable requirements of 3.5. The transparency of the handle shall permit the light flashes of the neon tube within the handle to be plainly visible.

3.14 Type VIII, bit set and handle. The type VIII bit set and handle may be procured as a complete set or each item may be procured individually as specified (see 6.1). The set shall include all items covered under type VIII, classes 1, 2, 3, and 4. All items shall conform to the requirements of 3.14.1 through 3.14.4. In order to assure interchangeability, the tang end of the bits and extension shall conform to the dimensions specified for type VIII, class 2 screwdrivers (see 3.14.2).

3.14.1 Class 1, plastic handle with clutch drive. The type VIII, class 1 handle shall be of plastic conforming to the applicable requirements of 3.5. The overall length of the handle shall be between 3-9/16 and 3-7/8 inches. The major diameter of the handle shall be between 31/32 and 1-1/16 inches. When a suitable bit is engaged with the inserted clutch drive, the handle shall be capable of withstanding a torsional moment of 75 inch-pounds when tested as specified in 4.4.2. The clutch drive shall satisfactorily engage, and hold in the working position, the bits and extension conforming to the applicable dimensions and tolerances specified herein (see 3.14.2, 3.14.3, and 3.14.4). The bits and extension shall not become disengaged from the clutch drive in lifting a weight of 2-1/2 pounds.

3.14.1.1 Type VIII, class 1 handles shall be similar to figure 20.

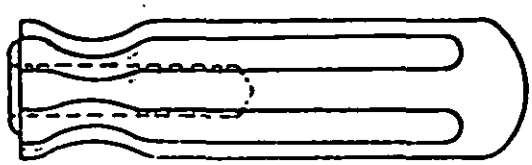
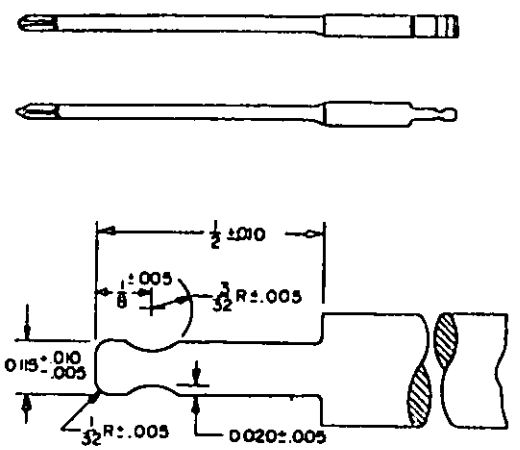


FIGURE 20. Type VIII, class 1 screwdriver, bit set, plastic handle with clutch drive.

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3.14.2 Class 2, bit, Phillips cross tip. Type VIII, class 2 bits shall conform to the applicable requirements of 3.10.3, shall be similar to figure 21 and conform to table XXII.



ENLARGED VIEW OF TANG END.

FIGURE 21. Bit and tang dimensions for type VIII, class 2 screwdriver, bit set, bit, Phillips cross tip.

TABLE XXII. *Type VIII, class 2 screwdriver, bit set, bit, Phillips cross tip*

Size	Over-all length +¼ −⅛	Diameter of point end ±0.010	Point size	Diameter of tang end ±0.010	Torsional moment (minimum)
Number	*Inches*	*Inch*	*Number*	*Inch*	*Inch-Pounds*
1	5	5/16	1	¼	25
2	5	¼	2	¼	60

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3.14.3 Class 3, bit, cabinet-type tip. Type VIII, class 3 bits shall conform to the applicable requirements of 3.4, 3.4.1, and 3.4.2. They shall be similar to figure 22 and conform to table XXIII.

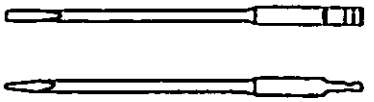


FIGURE 22. Type VIII, class 3 screwdriver, bit set, bit, cabinet type tip.

TABLE XXIII. Type VIII, class 3 screwdriver, bit set, bit, cabinet type tip

Size	Overall length +1/4 -1/8	Diameter of tip end ±1/64	Thickness of tip at end ±0.004	Diameter of tang end ±0.010	Torsional moment (Min.)
Number	Inches	Inch	Inch	Inch	Inch-Pounds
1	5-1/8	1/8	0.020	1/4	6
3	6-1/8	1/4	.037	1/4	40

3.14.4 Class 4, extension, bit. Type VIII, class 4 bit extension, for use with class 2 and 3 bits, shall be of high-grade steel and heat-treated to withstand a torsional moment of 60 inch-pounds. One end shall have a clutch drive for engaging the bits, and they shall not become disengaged in lifting a weight of 1-1/2 pounds. The other end shall have a tang conforming to the dimensions shown on figure 21. With the tang end inserted in a handle conforming to 3.14.1, and with a suitable bit engaged with the clutch drive end, the extension shall be subjected to the applicable tests of 4.4.2. The bit extension shall be similar to figure 23 and conform to table XXIV.

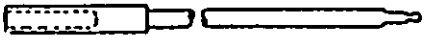


FIGURE 23. Type VIII, class 4, bit set, extension, bit.

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TABLE XXIV. *Type VIII, class 4, bit set, extension, bit*

Size	Over-all length ±1/8	Diameter of tang end ±0.010	Diameter of clutch drive end Max.	Torsional moment (minimum)
Number	*Inches*	*Inch*	*Inch*	*Inch-Pounds*
1	6-1/4	1/4	0.400	60

3.15 Type IX, flat tip - cross tip, close quarter. Type IX screwdrivers shall consist essentially of a plastic handle, and a reversible bit with a cabinet-type flat tip on one end, and a No. 2 Phillips cross tip on the other (see 3.15.2). A steel insert shall be molded or otherwise permanently secured in the handle for driving the bit. The surfaces of the steel insert and bit shall be coated as specified in 3.4.1. The handle shall be of plastic conforming to the applicable requirements of 3.5. The handle shall be capable of withstanding a torsional moment of 60 inch-pounds.

3.15.1 Handle insert. A steel insert having a hexagon socket shall be permanently secured in the plastic handle to a depth of 1-1/4 inches, ±3/16 inch. The hexagon section of the insert shall have an inside bearing length of at least 1/2 inch, and the dimensions across the flats shall be suitable for the hexagon section of the blade. A positive stop shall be located on the inside of the insert at a suitable distance from the outer end to limit the depth of engaging the hexagon section of the reversible bit, and to insure that the bit will not bottom in the hole. The hexagon section may have spring prongs for gripping the hexagon section of the reversible bit. The dimension from the bottom of the hole to the butt end of the handle shall be 1/8 inch minimum.

3.15.2 Reversible bit. The bit shall have a hexagon section integral with or permanently fastened to the middle portion of the bit. The hexagon section shall have a spring-loaded ball, or other suitable means, to prevent the handle and bit from becoming disengaged in lifting a weight of 1-1/2 pounds. One end of the bit shall have a No. 2 Phillips cross-tip point, the other end shall have a cabinet-type flat tip.

3.15.2.1 Phillips cross tip. The Phillips cross tip shall conform to the applicable requirements of 3.10.3 and 3.10.3.1, shall be capable of withstanding a torsional moment of 60 inch-pounds, and the applicable tests specified in 4.4.

3.15.2.2 Cabinet-type flat tip. The flat tip shall conform to the applicable requirements of 3.8, and be capable of withstanding a torsional moment of 40 inch-pounds. The tip shall be capable of withstanding the applicable tests specified in 4.4.

3.15.3 Hardness of hexagon sections. The hexagon socket of the handle, and the hexagon section of the bit, shall show a hardness of 48 to 58 on the Rockwell C scale, or shall be casehardened to show a reading of 84 to 89 on the Rockwell 15-N scale.

3.15.4 Type IX handle and bit shall be similar to, and conform to the dimensions and tolerances shown on either figure 24 or 25.

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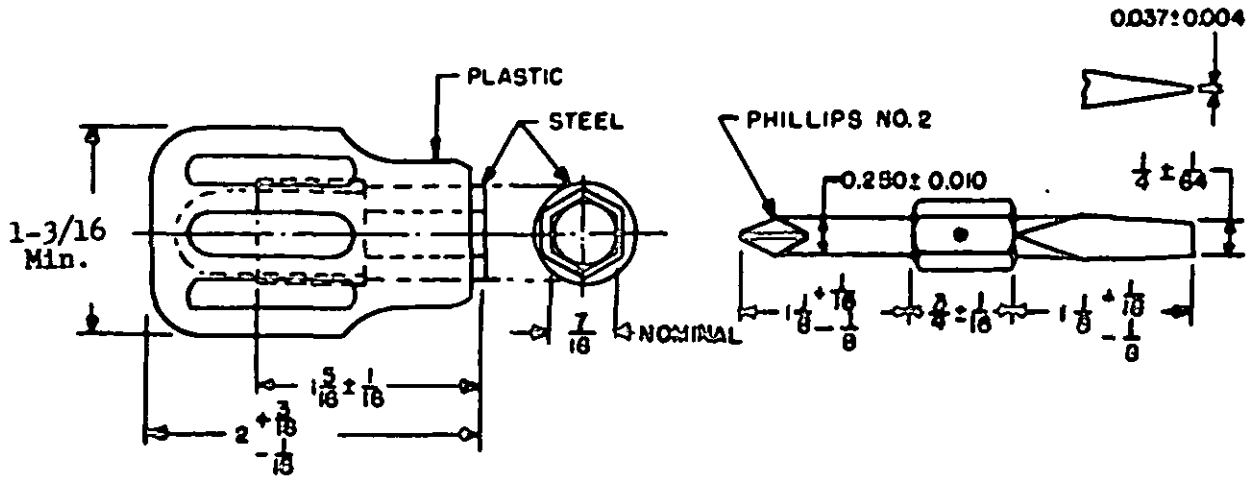


FIGURE 24. Type IX screwdriver, cabinet-type flat tip - cross tip, close quarter.

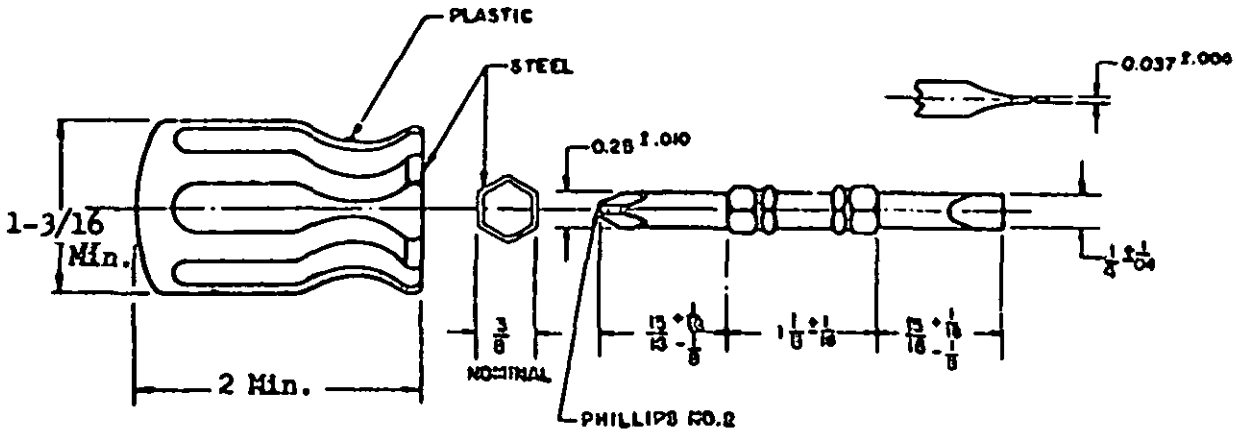


FIGURE 25. Type IX screwdriver, cabinet-type flat tip - cross tip, close quarter, optional design.

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3.16 Type X, clutch tip. The type X screwdrivers shall be similar to figures 26 and 27 and of the dimensions specified in tables XXV and XXVI. It shall consist of a blade and a plastic handle. The blade shall be in accordance with 3.4 through 3.4.2, and the handle shall be in accordance with 3.5.

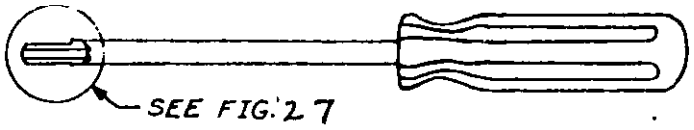


FIGURE 26. Clutch tip screwdriver.

TABLE XXV. Clutch tip screwdrivers

Clutch size	Blade length ±1/4	Handle dimensions		Torsional moment (minimum)
		*Diameter or distance across flats (min.)	Length (min.)	
Inch	Inches	Inch	Inches	Inch-pounds
3/32	3	11/16	1-3/4	8
1/8	3	11/16	1-3/4	20
5/32	4	13/16	2-5/8	30
3/16	4	13/16	3-1/4	40
1/4	6	15/16	3-1/2	70
9/32	6	1-1/8	3-1/2	80
5/16	6	1-1/4	3-1/2	90

*When handles with inserts are furnished, the diameter of the handles shall not be less than 1-9/32.

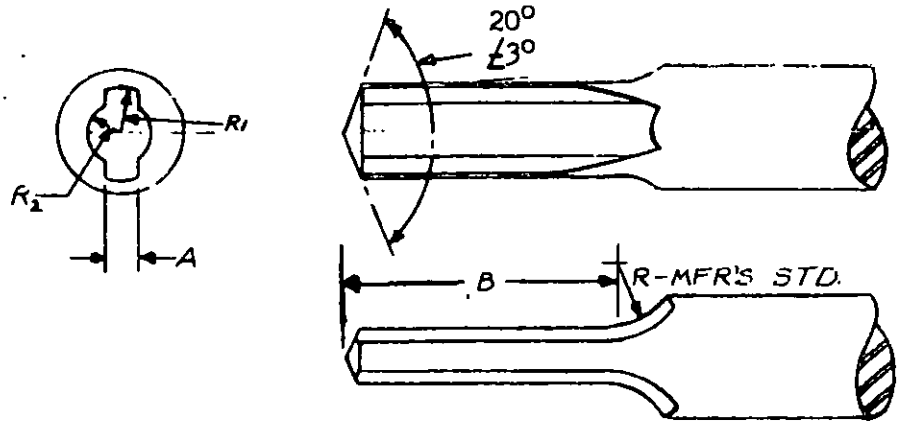


FIGURE 27. Clutch tip.

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TABLE XXVI. Dimensional requirements for clutch tips

Clutch size	A		B	R1		R2	
	Min.	Max.	Min.	Min.	Max.	Min.	Max.
Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch
3/32	.030	.034	1/16*	.0415	.0435	.0275	.0295
1/8	.042	.046	1/8	.0575	.0595	.0375	.0395
5/32	.054	.059	1/4	.0765	.0785	.0495	.0515
3/16	.064	.069	1/4	.0885	.0905	.0575	.0595
1/4	.088	.093	1/4	.1195	.1215	.0768	.0788
9/32	.100	.105	1/2	.1345	.1365	.0880	.0900
5/16	.110	.115	1/2	.1495	.1515	.0975	.0995

* Flat on tip

3.17 Workmanship. The screwdrivers shall conform to the requirements in this specification. They shall be free from rust and defects which may impair their serviceability, durability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table XXVII.

TABLE XXVII. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level (AQL)	AQL expressed in terms of	Reference
Visual examination	One each	II	2.5	Defects per 100 units	4.3.1
Dimensional examination	One each	S-3	4.00	Defects per 100 units	4.3.2
Testing	One each	S-3	4.00	Percent defective	4.4
Examination of preparation for delivery	One container	S-2	4.00	Defects per 100 units	4.4.12

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4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination.. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.11, as applicable.

4.4.1 Hardness. The Rockwell hardness tests shall be conducted in accordance with ASTM E18. Care shall be exercised to assure that the flat side of the blade or bit under test is normally at right angles to the axis of the penetrator of the testing machine.

4.4.2 Torsional moment test. The sample screwdriver shall be placed in an oven with the thermostat set for 125° F. The screwdriver shall remain in the oven for one hour. Then it shall be removed and within one minute the torsional moment test shall be conducted in the following manner: The blade end of each sample item under test shall be located in the applicable test block shown in table XXVIII. The stress shall be applied by a force acting at or near the middle of the natural grip of the handle, the force to act substantially at right angles to the axis of the screwdriver or screw starter. The stress shall be applied with the blade end held securely in the fixed slot while force is applied to an arm, bar, or clamp secured to the handle; or the stress may be applied by a force applied to the handle, under which the screwdriver or screw starter will be required to lift a dead weight attached to a lever arm, or in an equally satisfactory manner which will subject the tool to a known torsional stress from the handle end while the blade end is located in the test block. The force shall be applied by means of dead weights, through a spring-balance, or other suitable load-measuring device. The torsional moment shall be applied in right- and left-hand directions alternately for type V, classes 1 and 2 screwdrivers. The torsional moment shall be computed by multiplying the force in pounds by the arm in inches (15-pound load applied 10 inches from axis of tool = 150 inch-pounds, and so forth). The arm is the distance from the axis of the tool to the direction of the force at right angles to the latter. Figure 28 illustrates a device and the method employed in conducting torsional moment tests on screwdrivers or screw starters. It is permissible to support the blade at or near the junction of the blade and handle in order to maintain the blade in a suitable position for test. The screwdriver or screw starter may be held from endwise movement by a stop placed at handle end, provided the stop does not exert pressure on the handle end when assembled and ready for testing.

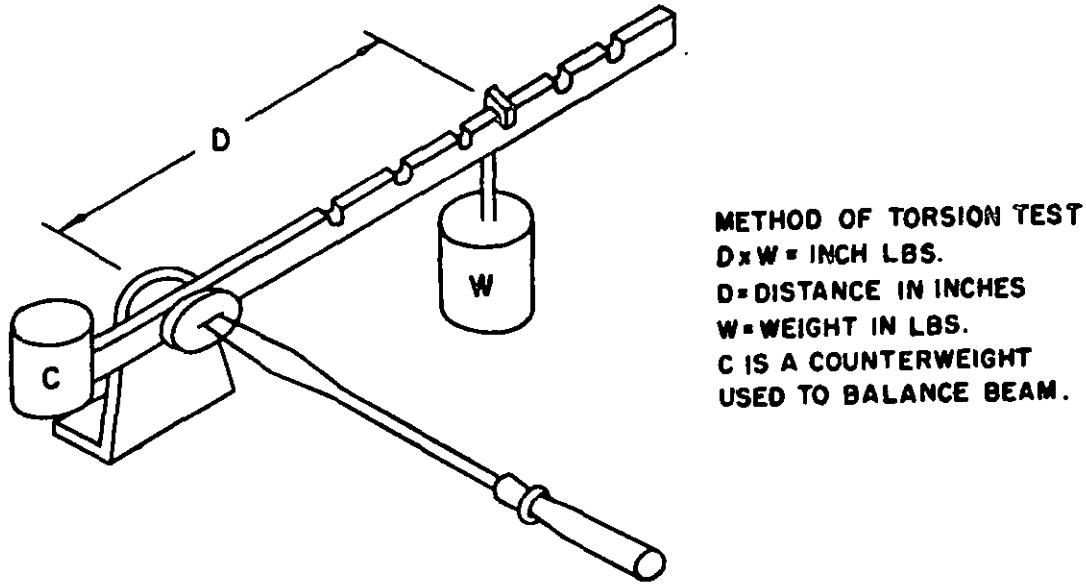


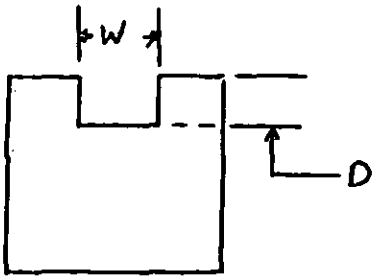
FIGURE 28. Torsional moment test.

***Note.*—The above method is not intended to restrict the manner in which the required tests shall be taken.**

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TABLE XXVIII. Test blocks for torsion test

Nominal tip thickness	Test slot width* (W)	Test slot depth* (D)	Test block designation
Inch	Inch	Inch	
0.015	0.025	0.023	A
.020	.030	.028	B
.025	.035	.033	C
.030	.042	.038	D
.032	.044	.040	E
.037	.050	.045	F
.042	.056	.050	G
.050	.064	.058	H
.058	.074	.066	I
.070	.086	.078	J



*Tolerance: ±0.001 inch.

4.4.2.1 Phillips and Reed and Prince crosstip screwdrivers and bits. In lieu of slots when testing these screwdrivers, the blade end of each sample under test shall be placed in a die having in its center the corresponding size of Phillips or Reed and Prince recess. The recesses shall be of such depth that the wing spread at the surface shall be as follows:

Phillips point size number	Wing spread Inch
0	0.090
1	.142
2	.233
3	.386
4	.486

Reed and Prince blade size Inch	Wing spread Inch
0.187	0.187
.250	.250
.312	.312
.375	.375

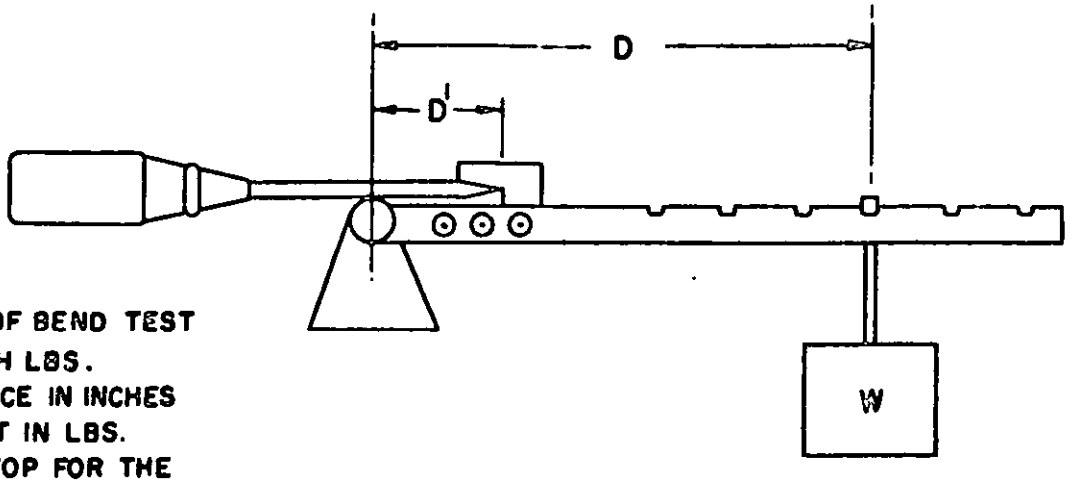
4.4.2.2 Type V, class 2 screwdrivers, ratchet, double spiral, automatic return. The torsional moment test shall be applied in alternate directions with the spindle locked in position, all in, and all the way out. In this test, the bits shall be assembled in position and the torsional moments shall be applied in accordance with 4.4.2 and tables XVI, XVII, XVIII, and XIX as applicable.

4.4.2.3 Type X, clutch tip. The torsional moment test shall be conducted in the following manner: The blade end, of each sample under test, shall be located in a test mandrel or screw conforming to the corresponding clutch tip recess. The load shall be applied by a force acting at or near the middle of the natural grip of the handle.

4.4.2.4 Torsional moment. When tested as specified in 4.4.2 through 4.4.2.3, the screwdrivers or screw starters shall not fail nor show visual indications of weakness of any nature, such as turning of the blade in ferrule or handle, splitting or breaking of the handle, slipping of the bits, twisting, distortion, or damage of any part of the tool or operating mechanism.

4.4.3 Bending moment test. The bending moment test shall be conducted in a manner similar to that shown on figure 29. In this test, the force shall be applied near the middle of the natural grip of the handle, the force to act substantially at right angles to the axis of the screwdriver. The load on the point end shall be supported not more than 1/8 inch from the end of the point of the blade. A spring-balance or other suitable load-measuring device may be used in lieu of a dead weight incident to applying the inch-pounds bending test. When tested in this manner, the blades shall not fail, show any visual permanent set, or loosen in the handle.

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METHOD OF BEND TEST
D x W = INCH LBS.
D = DISTANCE IN INCHES
W = WEIGHT IN LBS.
D' IS A STOP FOR THE SCREWDRIVER POINT AND IS LOCATED AT A DISTANCE EQUAL TO $\frac{1}{2}$ THE BLADE LENGTH.

FIGURE 29. Bending moment test.

Note.—The above method is not intended to restrict the manner in which the required tests shall be taken.

4.4.4 Compression load-test. Type V, class 2, spiral ratchet screwdrivers shall be subjected to the compression load specified in table XIV with the double spiral spindle all in, half out, and all the way out. The load shall be applied from the handle end by a pry bar with the bit end resting on the platform of a conventional style of platform scale, or in an equally satisfactory manner. When subjected to this test, spiral ratchet screwdrivers shall not fail, nor show any indications of weakness of any nature, such as twisting, distortion to the screwdrivers, or damage to the operating mechanism.

4.4.5 Blade-end toughness test. The flat tip end of the blade shall be inserted in a vise to a depth equal to four times the tip thickness of the blade, and the handle pulled slowly as a lever until either a definite permanent bend or a fracture at the blade end results. If fracture occurs, the pieces shall be refitted and the blade end shall show that a definite noticeable permanent bend had occured prior to fracture. If no permanent bend of a fractured blade end is apparent, the blade end is brittle and the screwdriver shall be considered to have failed the toughness test requirements.

4.4.6 Plastic handle tests. All plastic handled screwdrivers and screwdrivers with cushion grips shall resist the following test without damage: Immerse the handle in brake fluid conforming to VV-B-680, remove and let stand for 72 hours and test to applicable torque load. Repeat the test (using a new screwdriver) with oil conforming to either MIL-L-7808 or MIL-L-23699. The original hardness of cushion grips shall be 60 to 75 Shore A, and the subsequent change after the brake fluid and oil tests shall not be greater than 80 Shore A.

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4.4.7 Impact tests.

4.4.7.1 Impact test at room temperature. The blade end of each applicable screwdriver shall be mounted vertically in a vise affixed to the base of a suitable falling weight impact device. The blade tip shall rest on a solid surface to insure that the blade does not move vertically in the vise. The weight shall be 15 pounds, cylindrical in shape, and shall be dropped through a piece of seamless tubing slightly larger than the diameter of the weight in order to insure that the full force of the falling weight will be acting normal to the striking surface. The screwdriver handle shall not break, crack, or significantly distort, nor shall the blade penetrate more than 5/8 inch into the handle when the weight has been dropped ten times from the heights shown in table XXIX except close quarter screwdrivers (see 4.4.7.2). A 3/4 inch penetration in lieu of 5/8 inch shall apply for blade diameters 3/16 inch and smaller. Significantly distort (for purpose of this test) means an increase of at least 5 percent in the handle diameter, either of a uniform or irregular bulge. In conducting this test, care shall be taken that the impact energy will not be expended in flexing of the blade or in driving the screwdriver tip into the surface on which it rests. The blade shall be shortened or blunted, if necessary, to insure a proper test.

TABLE XXIX. Height of drop of 15 pound weight for impact tests

Blade diameter (stock size)	Height of drop of 15 pound weight for impact tests
	At room temperature
Inch	Inches
1/8	1-1/2
9/64	1-1/2
5/32	4
3/16	6
7/32	8
1/4	10
9/32	12
5/16	15
11/32	17
3/8 and over	20

4.4.7.2 Close quarter screwdrivers. For close quarter screwdrivers only, the height of drop shall be decreased to one-half the distance specified in table XXIX and the 5/8 inch maximum penetration into the handle shall be reduced to 7/16 inch.

4.4.8 Screw gripping tests for type I, class 6, type III, and type VI, class 3 screwdrivers and screw starters.

4.4.8.1 The screw starters shall be connected, by engaging the gripping mechanism, to a machine screw of a size for which the point thickness of the blade is suited. The screw shall be driven at least three full turns into a nut of proper thread size. The sample shall be rejected if the gripping mechanism becomes disengaged in the process of starting or driving the screw.

4.4.8.2 The screw starter shall be connected to a stock screw of proper size for the screw starter point by means of the gripping mechanism. The sample shall be rejected if the screw becomes disengaged from the screw starter in lifting a weight of one and one-half pounds, except for the miniature, type III, which shall be required to lift a weight of 3/4 pound.

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4.4.9 Female drive insert. The screwdriver with a female drive insert shall be mounted on a corresponding male drive, and shall retain the male drive with a retention force of no less than four pounds, and shall also withstand a torque load equal to the torque load for the screwdriver without slipping, turning, or breaking.

4.4.10 Surface finish test. Each of the sample screwdrivers shall be tested for surface finish in accordance with ANSI B46.1 to determine compliance with the surface finish requirements specified (see 3.4.1).

4.4.11 Magnetic test. The type I, class 8, style 1, magnetized tip, screwdrivers shall pick up small ferrous screws when the screwdriver blade is passed over them. The style 2, nonmagnetized tip, screwdrivers shall not pick up the screws when subjected to the same test.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine whether the preservation, packaging, packing, and marking comply with the requirements of section 5 of this specification.

5. *PREPARATION FOR DELIVERY*

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of packaging shall be A, B, or C. The level of packing shall be A, B, or C, and the marking shall be civil or military as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, design, and style, as applicable (see 1.2).
(c) Size (see applicable tables).
(d) Handles with 3/8 inch female drive (see 3.6).
(e) Whether a cushion grip is to be furnished (see 3.7).
(f) Whether additional blades, or blades only, shall be furnished for type I, class 7 screwdrivers (see 3.8.6).
(g) Whether style 1 (magnetized) or style 2 (nonmagnetized) pocket-type screwdriver is required (see 3.8.7).
(h) Separate bits when required for type V, class 2 screwdrivers (see 3.11.2.1 and 3.11.2.3).
(i) Phillips type bits and bit number(s) when required for type V, class 2 screwdrivers (see 3.11.2.2).
(j) Reed and Prince type bits and bit number(s) when required for type V, class 2 screwdrivers (see 3.11.2.3).
(k) Size bit required (see 3.12.3).
(l) Whether type VIII screwdrivers should be furnished in sets or as individual items (see 3.14).
(m) Level of packaging, packing, and type of marking required (see 5.1).

6.2 Types and classes of screwdrivers covered in this specification corresponding to those covered in Federal Specification GGG-S-121d are as follows:

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GGG-S-121E	GGG-S-121D
Type I - Flat tip.	Type I - Flat tip.
Class 1 - Cabinet, straight sides.	Class 1 - Cabinet, straight sides, plastic handle.
Design A - Plain.	Design A - Plain.
Combined in table I.	Style 1 - Normal duty. Style 2 - Light duty. Style 3 - Extra-light duty.
Deleted.	Design B - Bolster, forged.
Deleted.	Style 1 - Normal duty.
Deleted.	Style 2 - Light duty.
Deleted.	Style 3 - Extra-light duty.
Deleted.	Class 2 - Cabinet, straight sides, wood handle.
Deleted.	Design A - Plain.
Deleted.	Style 1 - Normal duty.
Deleted.	Style 2 - Light duty.
Deleted.	Style 3 - Extra-light duty.
Deleted.	Design B - Bolster, attached or forged.
Deleted.	Style 1 - Normal duty.
Deleted.	Style 2 - Light duty.
Deleted.	Style 3 - Extra-light duty.
Class 3 - Close quarter.	Class 3 - Close quarter, plastic handle.
Class 4 - Heavy duty, flared sides.	Class 4 - Heavy duty, flared sides.
Deleted.	Style 1 - Blade extending through wood handle.
Deleted.	Shape a. - Round blade.
Deleted.	Shape b. - Square blade.
Style 2 - Steel handle with wood inserts.	Style 2 - Steel handle with wood inserts.
Deleted.	Shape a. - Round blade.
Deleted.	Shape b. - Square blade, extra heavy duty.
Class 5 - General purpose, flared sides.	Class 5 - General purpose, flared sides.
Deleted.	Style 1 - Plastic handle.
Design A - Plain, medium heavy duty.	Design A - Plain, medium heavy duty.
Deleted.	Shape a. - Round blade.
Deleted.	Shape b. - Square blade.
Design B - Bolster, forged, with hexagon wrench-grip.	Design B - Bolster, forged, heavy duty.
Deleted.	Shape a. - Round blade.
Deleted.	Shape b. - Square blade, or round blade with hexagon-wrench grip.
Deleted.	Style 2 - Wood handle.
Deleted.	Design A - Plain.
Deleted.	Shape a. - Round blade.
Deleted.	Shape b. - Square blade.
Deleted.	Design B - Bolster attached or forged.
Class 6 - Flared sides with screw starter, external gripping.	Class 6 - Flared sides with screw starter, external gripping, plastic handle.
Class 7 - Pocket, removable blade.	Class 7 - Pocket, removable blade.
Class 8 - Pocket, clip.	Class 8 - Pocket, clip, plastic handle.
Style 1 - Magnetized tip.	Style 1 - Magnetized tip.
Style 2 - Nonmagnetized tip.	Style 2 - Nonmagnetized tip.
Deleted. See GGG-S-1808.	Type II - Jewelers'.
Deleted.	Class 1 - Swivel knob, interchangeable blades.
Deleted.	Class 2 - Nonmetallic handle, nonremovable blade.

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GGG-S-121E	GGG-S-121D
Type III - Screw starters, hand. Deleted. Deleted. Deleted.	Type III - Screw starters, hand. Class 1 - Center rotating tip, plastic handle. Class 2 - Center rotating tip, metal handle. Class 3 - Sliding action, plastic handle.
Type IV - Offset. Class 1 - Double offset, single flat tip each end. Class 2 - Four-way offset, double flat tip each end. Class 3 - Phillips cross tip each end. Style 1 - Double offset. Style 2 - Single offset. Deleted.	Type IV - Offset. Class 1 - Double offset, single flat tip each end. Class 2 - Four-way offset, double flat tip each end. Class 3 - Phillips cross tip each end. Style 1 - Double offset. Style 2 - Single offset. Class 4 - Double offset, Reed and Prince cross tip each end.
Type V - Ratchet. Class 1 - Common, straight. Class 2 - Double spiral, automatic return. Style 1 - Light duty. Style 2 - Medium duty. Style 3 - Heavy duty.	Type V - Ratchet. Class 1 - Common, straight. Class 2 - Double spiral, automatic return. Style 1 - Light duty. Style 2 - Medium duty. Style 3 - Heavy duty.
Type VI - Cross tip, straight. Class 1 - Phillips Deleted. Deleted. Class 2 - Reed and Prince. Class 3 - Phillips, with screw starter, external gripping, with cushion grip.	Type VI - Cross tip, straight. Class 1 - Phillips. Style 1 - Plastic handle. Style 2 - Wood handle. Class 2 - Reed and Prince.
Type VII - Screwdriver, spark testing.	Type VII - Screwdriver, spark testing.
Type VIII - Bit set and handle. Class 1 - Plastic handle with clutch drive. Class 2 - Bit, Phillips cross tip. Class 3 - Bit, cabinet-type tip. Class 4 - Extension, bit.	Type VIII - Bit set and handle. Class 1 - Plastic handle with clutch drive. Class 2 - Bit, Phillips cross tip. Class 3 - Bit, cabinet-type tip. Class 4 - Extension, bit.
Type IX - Flat tip - cross tip, close quarter.	Type IX - Flat tip - cross tip, close quarter.
Type X - Clutch tip.	Interim Federal Specification GGG-S-00122A(GSA-FSS).

MILITARY CUSTODIANS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

User activities:

Army - GL
Air Force - 84

Preparing activity:

GSA-FSS

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